(Non)commutative ball maps

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Let H^2 be the space of square-summable power series (these are holomorphic in $\mathbb{D} = \{|z| < 1\}$).

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Theorem[Beurling's theorem in finite codimension]
Let $\mathcal{M}$ be invariant, $\dim \mathcal{M}^\perp = n < \infty$

- There is a rational function φ, regular across the boundary of $\mathbb{D}$ such that $P_{\mathcal{M}} = M_\varphi M_\varphi^*$ and

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- the <u>full Fock space</u> (the "noncommutative" or "free" setting) (Beurling-type theorems due to Arias/Popescu 1995, Davidson/Pitts 1998)

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What we want: finite codimension multiplier-invariant subspaces should be ranges of "rational", "inner" functions, and maybe more details...

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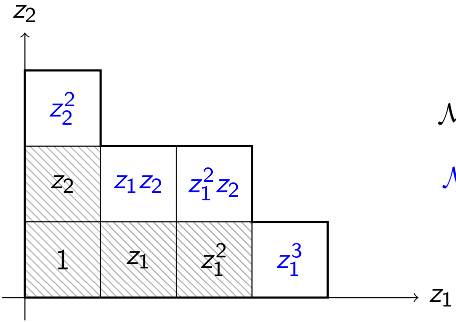
- say a closed subspace $\mathcal{M} \subset H^2_g$ is invariant if $\varphi\mathcal{M} \subset \mathcal{M}$ for all multipliers (equivalently, just for coordinate functions z_j)

Let $\mathcal{M} \subset H^2_g$ be an invariant subspace. Define

$$\mathcal{M}_\partial = \mathrm{span}\{1, z_j u : u \in \mathcal{M}^\perp\} \ominus \mathcal{M}^\perp$$

The ball codimension of $\mathcal{M}$ is

$$m := \dim \mathcal{M}_\partial$$



$\mathcal{M}^\perp = \mathrm{span}\{1,\ z_1,\ z_2,\ z_1^2\}$

$\mathcal{M}_\partial = \mathrm{span}\{z_2^2,\ z_1 z_2,\ z_1^2 z_2,\ z_1^3\}$

$m = 4$

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$$M \geq m \quad \text{and} \quad \psi = \varphi V^*$$

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So when $g = 1$, ball codimension is always 1, and φ is a finite Blaschke product

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Theorem 2 (zero-based invariant subspaces) If $\Lambda = \{\lambda_1, \ldots, \lambda_n\}$ are distinct points and $\mathcal{M}$ is the space of functions vanishing on Λ, then

$$\dim \mathcal{M}_\partial = n(g-1) + 1 \quad \text{(the upper bound is attained)}$$

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Problem: Give a satisfying account of the "gaps" (compare SCV works of D'Angelo et al.)

hydration break

pause fraîcheur

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where $Z^w = Z_{i_1} Z_{i_2} \cdots Z_{i_n}$ for $w = i_1 i_2 \cdots i_n$

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  are isometries with orthogonal ranges...also right shifts R_j
- There is a Beurling-style theorem for L (or R) invariant subspaces $\mathcal{M} \subset \mathcal{F}_g^2$ (Arias/Popescu '95, Davidson/Pitts '98)

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- choice of free point $\mathbf{Y}$ is unique up to a similarity

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N.B. no "ball codimension" in the free setting (or, it's always the maximal value $n(g-1)+1$), and so no "gap conditions"